

**First Lucid Hydra board**

MSI Big Bang Fuzion board was the first board to support the Lucid Hydra 200 chip

**NVIDIA Optimus**

NVIDIA Optimus is an optimization technology that switches off the GPU power when not in use and directs the graphics commands to integrated graphics

# Green and Red Armies

Gaming enthusiasts have always tried to crank up their game settings to get the most out of their games. Some of them even run a multi-GPU set up to see the scaling in frame rates. NVIDIA's Scalable Link Interface (SLI) and AMD's CrossfireX technology allows you to combine the power of two similarly specced cards, with Crossfire allowing you to use two different cards. The rivalry between these two companies has ensured that these technologies work on certain motherboards which support them. For instance, AMD chipsets do not support SLI out of the box, you have to find hacks to force SLI on motherboards with AMD chipsets. With such politics behind SLI and Crossfire setups, which essentially use their own GPUs, if someone told you that there is a company which allows you to run an AMD card and an NVIDIA card on the same motherboard, you will most likely be taken aback.

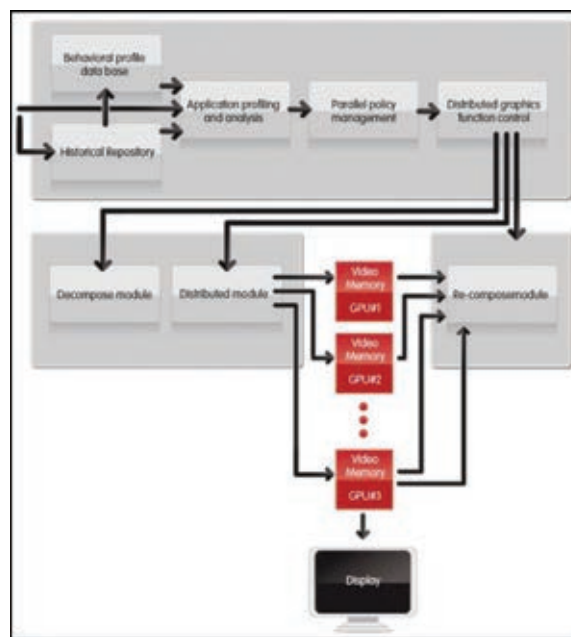
Lucid Logix is the name of the firm that first proposed a technology which allows a user to use multiple GPU set up in a variety of combinations including running an AMD and NVIDIA card alongside each other on the same board. Lucid is

## UNITE

We get behind the technology of the system that allows you to run an AMD as well as an NVIDIA card off the same motherboard

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Block diagram of the Lucid Hydra chip showing breaking up of tasks and assignment to different GPUs

an Israeli company which first showcased its Hydra Engine technology in September 2008 at the Intel Developers Forum. Back then, they demonstrated SLI on two different NVIDIA cards. The product has evolved and the latest Lucid Hydra 200 chip found on motherboards today, can support AMD and NVIDIA cards on the same board.

### How multi-GPU works?

Before jumping to how the Hydra technology works, a quick primer on the method in which SLI and Crossfire works.

**Split Frame Rendering (SFR) :** In this technique a frame is broken into two halves, with each GPU working on one half. In the end the two frames get combined to form one frame. In a Crossfire setup, dynamic load balancing is enabled in case of a set up involving two different cards (one faster, the other slower but from the same series). While load balancing, the faster card has to render a larger portion of the screen than the slower card, such that a frame is rendered at the same time irrespective of the speed of the card.

### Alternate Frame Rendering (AFR):

A more popular frame rendering technique used by both AMD and NVIDIA in which each GPU handles one complete frame. The catch being that one GPU works on the odd frame while the other works on the even frame.

**Supertiling:** This technique used in the Crossfire setup where in the entire frame is divided into 32x32 pixel blocks called Tiles, and each card handles a portion of the total available tiles. As Crossfire allows use of two cards from the same series, load balancing is used in this. Load balancing ensures that the card with a faster processor gets more tiles to render than the slower card.

### FIVE MODES OF LUCID HYDRA

| Mode                                               | Exclusive to LucidLogix | Example config                    |
|----------------------------------------------------|-------------------------|-----------------------------------|
| <b>N-mode (identical NVIDIA cards)</b>             | No                      | GeForce GTX 285 / GeForce GTX 285 |
| <b>N-mode (non identical NVIDIA cards)</b>         | Yes                     | GeForce GTX 285 / GeForce GTX 260 |
| <b>A-mode (identical AMD cards)</b>                | No                      | Radeon HD 5870 / Radeon HD 5870   |
| <b>A-mode (non identical AMD cards)</b>            | Yes                     | Radeon HD 4870 / Radeon HD 5870   |
| <b>X-mode (cards from different manufacturers)</b> | Yes                     | GeForce GTX 285 / Radeon HD 5870  |